

WALMART SALES PERFORMANCE DASHBOARD

Using Microsoft Power BI

Project Report

Submitted By

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Tool Used: Microsoft Power BI Desktop

1. Introduction

For this project, I developed an interactive Walmart Sales Performance Dashboard using Microsoft Power BI. The dashboard was created using Walmart's weekly sales data, which contains information related to stores, weekly sales, dates, fuel prices and holiday status.

The main purpose of the dashboard is to convert raw sales data into meaningful visual information that can be easily understood and analysed. Power BI was used to create KPI cards, charts and filters to examine Walmart's sales performance from different perspectives.

The dashboard provides an overall view of sales performance and allows users to analyse sales based on store, date, year and holiday status. It also helps in understanding changes in sales over time and examining the relationship between weekly sales and fuel prices.

2. Objectives

- To analyse Walmart's weekly sales performance across different stores.
- To create an interactive sales dashboard using Microsoft Power BI.
- To understand sales performance across the 45 Walmart stores.
- To analyse the trend of average sales over time.
- To compare sales during holiday and non-holiday periods.
- To examine the relationship between fuel prices and weekly sales.
- To provide important sales information through KPI cards.
- To enable users to filter and analyse the dashboard using interactive slicers.

3. Dataset Description

The dataset used for this project contains weekly sales information from Walmart stores. The dataset includes numerical and categorical fields that are useful for sales analysis.

- Store – Identifies the Walmart store, ranging from Store 1 to Store 45.
- Date – Represents the date or week for which the sales information was recorded.
- Weekly Sales – Represents the sales generated by a store during a particular week.
- Fuel Price – Represents the fuel price recorded during the corresponding period.
- Holiday Flag – Indicates whether the particular week was a holiday week or a non-holiday week.
- CPI – Represents the Consumer Price Index associated with the period.
- Unemployment – Represents the unemployment rate during the period.

4. Data Preparation

Before creating the dashboard, the dataset was imported into Microsoft Power BI Desktop and prepared for analysis.

- Imported the Walmart sales dataset into Power BI.
- Checked the available columns and their data types.
- Ensured that the Date field was recognized correctly for time-based analysis.
- Verified that Weekly Sales and Fuel Price were treated as numerical fields.
- Checked the Holiday Flag field for categorical analysis.

- Used the available fields to create suitable measures for the dashboard.
- Organised the data so that different visuals could interact with each other.

5. DAX Measures Used

DAX measures were used in Power BI to calculate important sales indicators and display them through KPI cards and visualisations.

Total Sales: Calculates the total value of weekly sales across the dataset.

Measure: Total Sales = SUM(Weekly_Sales)

Average Sales: Calculates the average weekly sales.

Measure: Average Sales = AVERAGE(Weekly_Sales)

Transaction Count: Represents the number of sales records/transactions available in the dataset.

Sales Count: Represents the number of stores included in the analysis.

6. Dashboard Components

The dashboard contains multiple visualisations, with each visual designed to present a specific aspect of Walmart's sales performance.

A. KPI Cards

The KPI cards provide a quick summary of the overall sales performance. The dashboard displays Total Sales of approximately ₹6.74 billion, Average Sales of approximately ₹1.05 million, a transaction count of approximately 6K, and a sales/store count of 45.

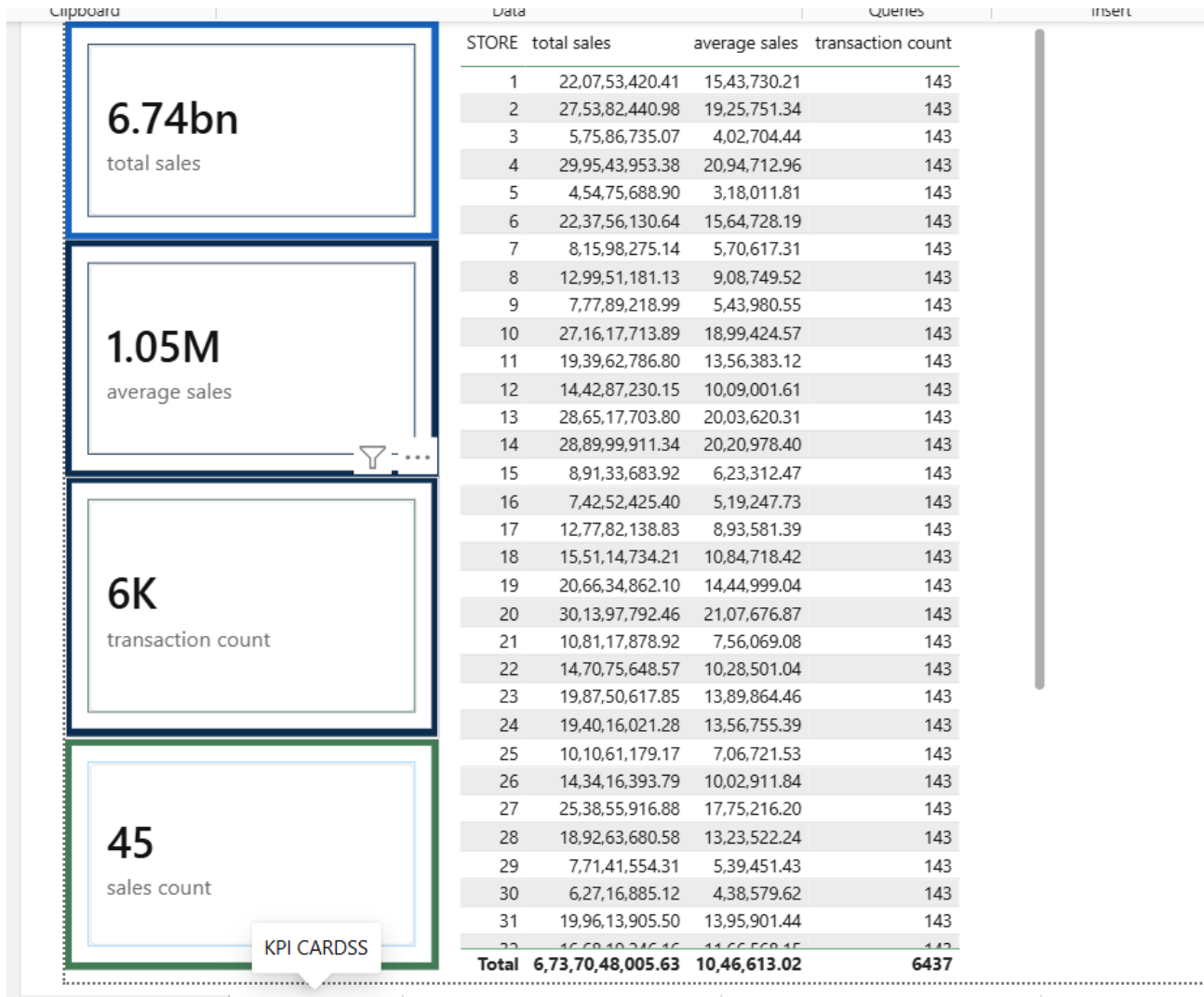


Figure 1: KPI cards and store-level sales data in Power BI.

B. Sum of Weekly Sales by Store

A column/bar chart is used to compare total weekly sales across the different Walmart stores. The chart covers all 45 stores and shows considerable variation in sales performance between stores. This visual helps identify stores with relatively higher or lower sales performance.

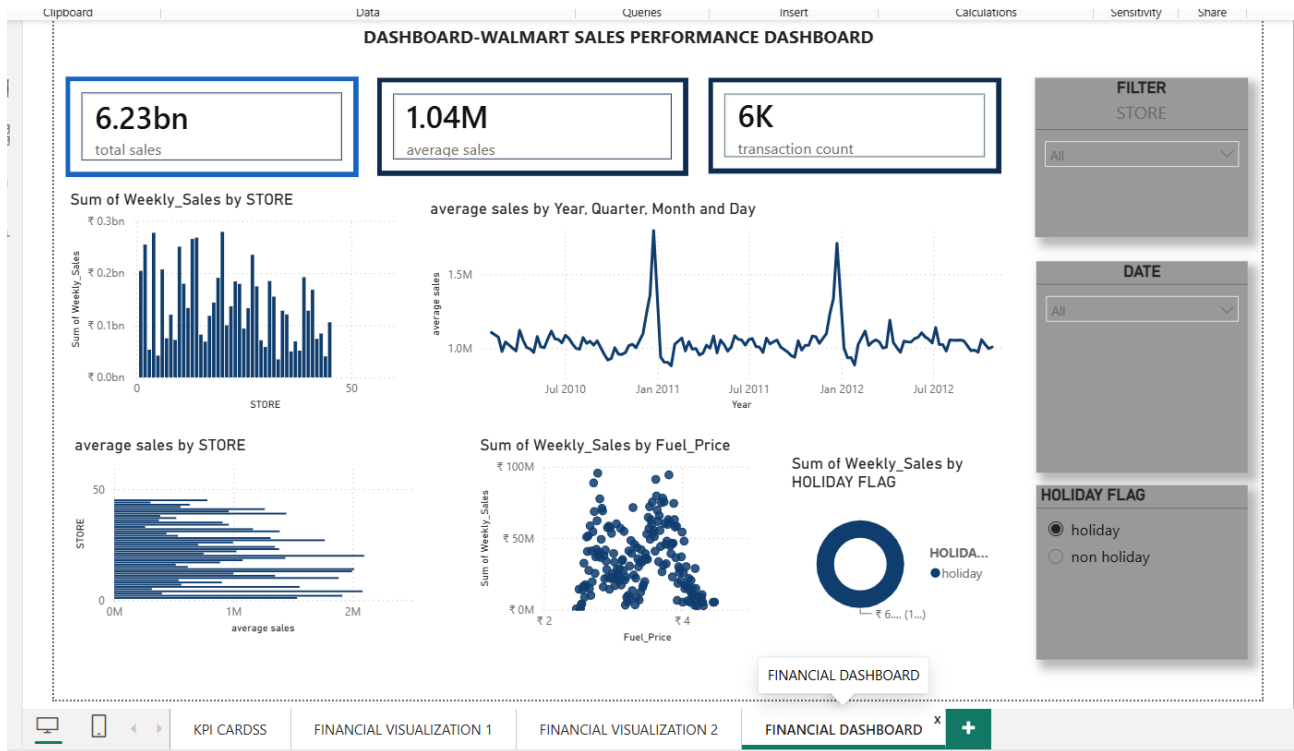


Figure 2: Final Walmart Sales Performance Dashboard showing store-wise sales.

C. Average Sales by Year, Quarter, Month and Day

A line chart is used to analyse the average sales trend over time. The chart shows fluctuations in average sales across the period covered by the dataset, including noticeable peaks during certain periods.

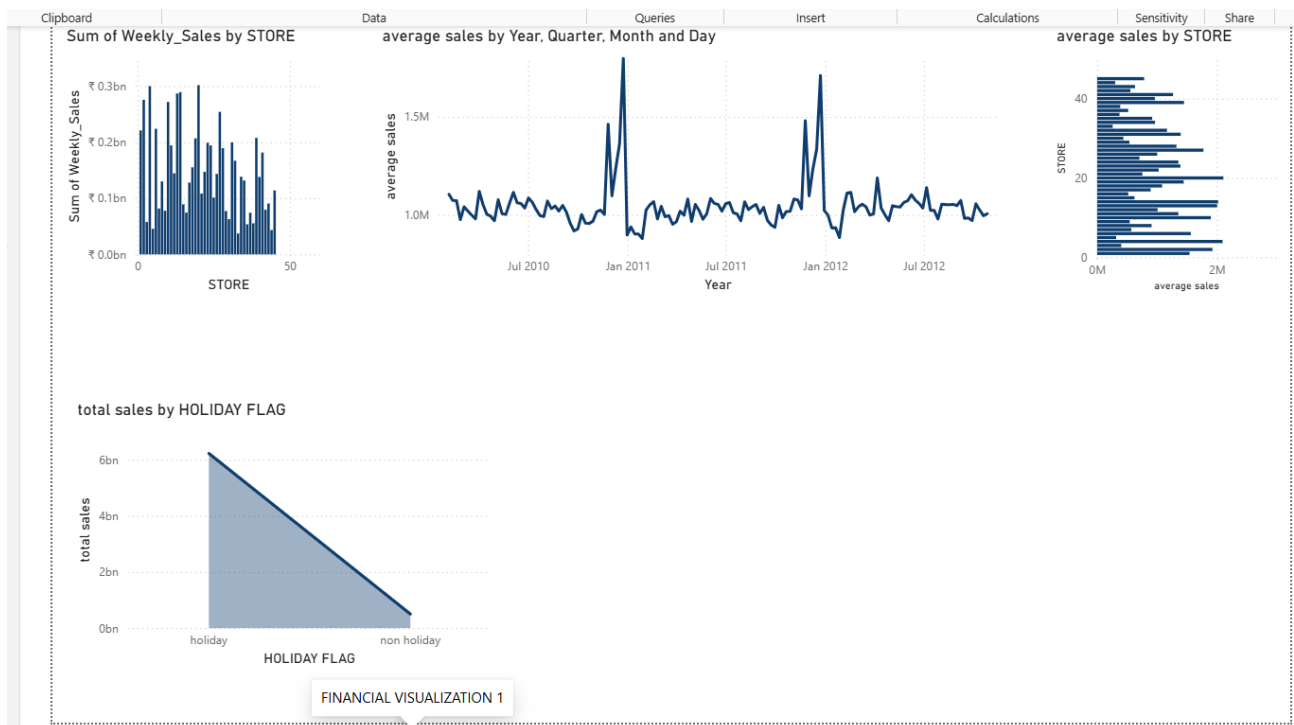


Figure 3: Financial Visualization 1 – average sales trend and store analysis.

D. Average Sales by Store

A horizontal bar chart displays the average sales for each Walmart store. This makes it easier to compare the average performance of individual stores and identify stores with relatively higher or lower average sales.

E. Fuel Price vs Weekly Sales

A scatter chart is used to analyse the relationship between Fuel Price and Weekly Sales. The X-axis represents Fuel Price, while the Y-axis represents Weekly Sales. The data points are widely scattered and do not show a strong clear linear relationship in the displayed data.

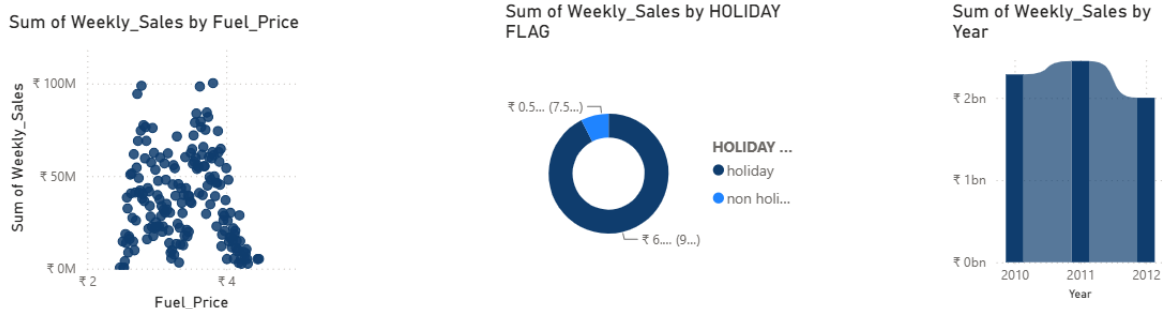


Figure 4: Financial Visualization 2 – fuel price, holiday and yearly sales analysis.

F. Weekly Sales by Holiday Flag

A donut chart is used to compare weekly sales during holiday and non-holiday periods. The chart divides total weekly sales into holiday and non-holiday categories. The non-holiday category contributes a larger share of total sales because there are more non-holiday weeks.

G. Total Weekly Sales by Holiday Flag

An additional visual compares total sales generated during holiday and non-holiday periods. This provides another view of how sales are distributed according to holiday status.

H. Sum of Weekly Sales by Year

The yearly sales visual compares total weekly sales across the years available in the dataset. The displayed analysis covers the period around 2010–2012 and allows a simple year-wise comparison.

7. Slicers Used

To make the dashboard interactive, slicers were added to allow users to filter the data.

- Store – Allows the user to select a particular Walmart store and analyse its sales performance.
- Date – Allows the user to narrow the analysis to a particular date or period.

- **Holiday Flag** – Allows the user to analyse holiday and non-holiday sales separately.

When a slicer is selected, the KPI cards and charts update according to the selected filter. This makes the dashboard more flexible than a static report.

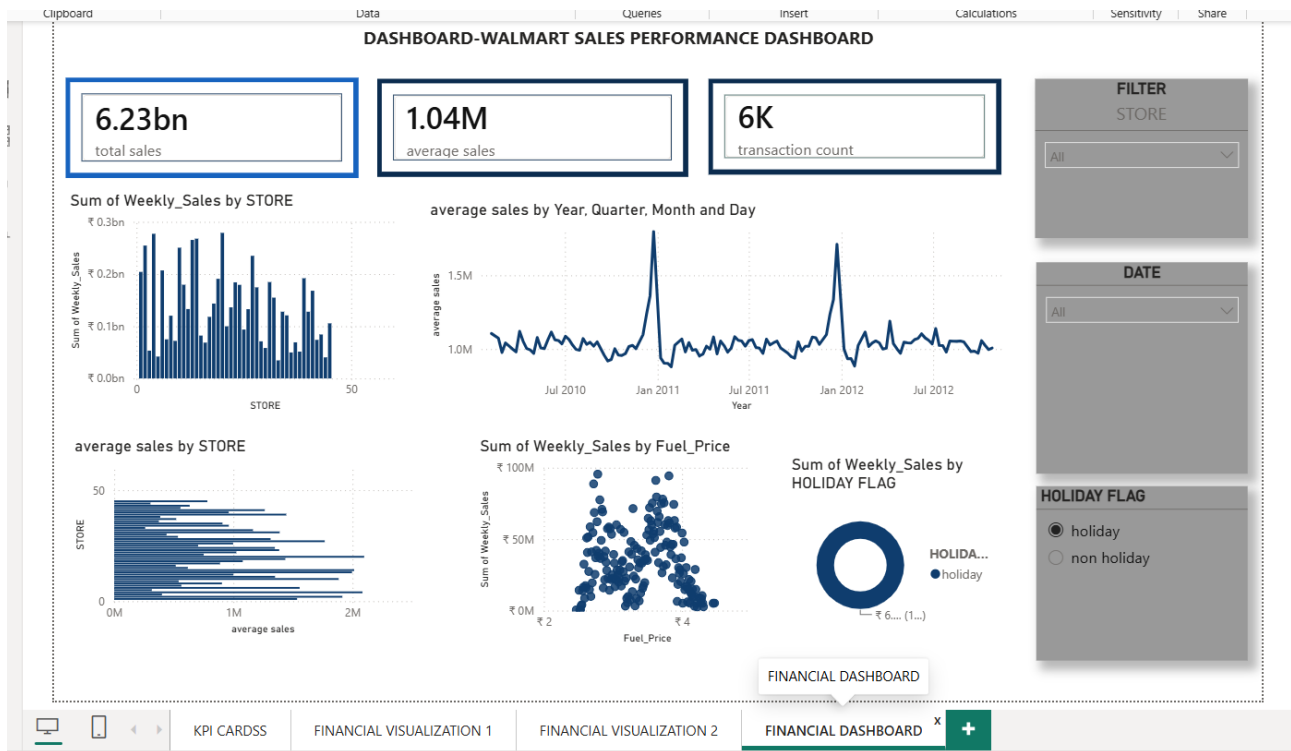


Figure 5: Interactive dashboard with Store, Date and Holiday Flag filters.

8. Key Findings

- The total sales across the complete dataset are approximately ₹6.74 billion.
- The overall average sales are approximately ₹1.05 million.
- The dataset covers 45 Walmart stores.
- Sales performance varies significantly between different stores.
- Some stores generate considerably higher total and average sales than others.
- Average sales fluctuate over time, with noticeable peaks during certain periods.
- Non-holiday weeks account for a larger portion of total sales because they occur much more frequently than holiday weeks.
- The scatter plot between fuel price and weekly sales does not show a strong direct relationship.
- Year-wise analysis shows changes in Walmart's sales performance across the period covered by the dataset.
- The interactive slicers make it possible to analyse individual stores and specific periods more effectively.

9. Conclusion

The Walmart Sales Performance Dashboard demonstrates how Microsoft Power BI can be used to transform raw sales data into an interactive and meaningful business intelligence dashboard.

The use of KPI cards provides a quick overview of sales performance, while charts such as store-wise sales, time-series analysis, holiday analysis and fuel-price comparison provide deeper insights.

The interactive slicers further improve the usefulness of the dashboard by allowing users to analyse specific stores, dates and holiday categories. Overall, this project helped in understanding how Power BI, data visualisation and basic DAX calculations can be combined to support sales analysis and business decision-making.

10. References

- Microsoft Power BI Desktop
- Walmart Sales Dataset
- Microsoft Power BI Documentation